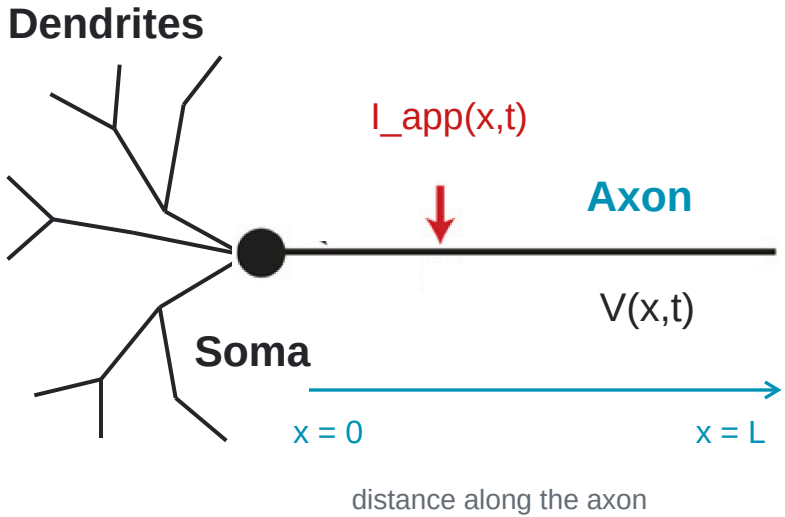
Distributed Hodgkin-Huxley model

An excitable membrane at each position along the axon



Voltage and local channel dynamics

$$C_M \partial_t V = \frac{r}{2R} \partial_{xx} V + I_{\text{app}} - I_{\text{ion}},$$

$$\begin{aligned} I_{\text{ion}} &= g_K n^4 (V - V_K) \\ &\quad + g_{\text{Na}} m^3 h (V - V_{\text{Na}}) \\ &\quad + g_l (V - V_l), \end{aligned}$$

$$\partial_t q = \frac{q_\infty(V) - q}{\tau_q(V)}, \quad q \in \{n, m, h\}.$$

The cable equation models the axon; dendrites and soma are shown for context.

Schematic: Brette (2015), Fig. 5C, adapted (CC BY). Model: Martinez et al. (2023), Eqs. (8)-(9).